4. Repeated Subtraction Method:

To find the HCF for the given two numbers say **m** and **n** we do the subtraction continuously until m and n are equal. For example,

Find the HCF of 144 and 120

STEP 1: Check whether m = n

Here, take **m** = 144 and **n** = 120

Check whether **m = n** or **m > n** or **m < n** Here **m > n (144 > 120).**

STEP 2: m > n perform m – n repeat the process till m = n or m < n perform n – m repeat the process till m = n

If m is greater than **n,** then we perform **m – n** and assign the result (the difference) as **m.** Again we check whether **m and n are equal or not** and repeat the process. If **m** is less than **n,** then we perform **n – m** and we assign the result (the difference) as **n**. Again we check whether **m and n are equal or not** and the process is repeated.

Subtract **120(n)** from **144(m)** till **m = n.**

First	144 – 120 = 24	Repeat	120 – 24 = 96	Repeat	96 – 24 = 72
Repeat	72 – 24 = 48	Repeat	48 – 24 = 24	Repeat	24 – 24 = 0

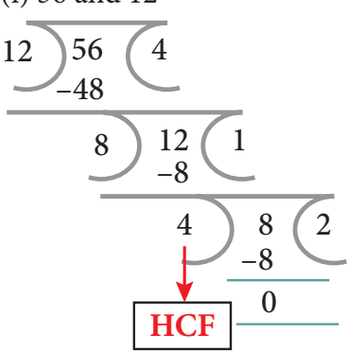
STEP 3: When m and n values are equal then that equal value will be the HCF (m, n).

Now m = n, Hence, we conclude that the **HCF of 144 and 120 is 24.**

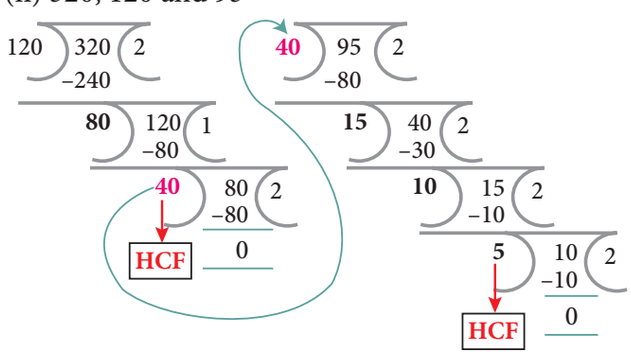
Comparing both the repeated division and repeated subtraction methods, in finding the HCF, we can conclude that the repeated subtraction, in one way is easier and gives the HCF faster that the repeated division and also that one would want to easily do subtraction rather than division. Isn't it ?

Exercise 7.2

1. Using repeated division method, find the HCF of the following:
   (i) 455 and 26 (ii) 392 and 256 (iii) 6765 and 610 (iv) 184, 230 and 276
2. Using repeated subtraction method, find the HCF of the following:
   (i) 42 and 70 (ii) 36 and 80 (iii) 280 and 420 (iv) 1014 and 654
3. Do the given problems by repeated subtraction method and verify the result.
   (i) 56 and 12



   (ii) 320, 120 and 95



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